INTERNATIONAL MONETARY FUND

BHUTAN

STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

May 30, 2014

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The Debt Sustainability Analysis (DSA) has been prepared jointly by IMF and World Bank staff, in consultation with the authorities, using the debt sustainability framework for low-income countries approved by the Boards of both institutions.

Based on the Joint Fund-Bank Debt Sustainability Framework for Low-Income Countries (LIC DSA), Bhutan's risk of external debt distress is moderate. Bhutan's rapid hydropower development is projected to lead to a substantial buildup of external debt in the medium-term, with debt ratios breaching most of the country-specific indicative thresholds for protracted periods of time. Also risks stem from volatile hydro-related debt service payments. Bhutan's debt situation is only expected to improve over the long run, reflecting significantly higher electricity exports when more hydropower projects come on stream. The staff's assessment is primarily based on unique mitigating factors, namely explicit guarantees from India that cover financial and construction risks for the hydropower projects, thus allowing for the exceptional treatment of hydropower loans from India as similar to FDI—i.e., non-debt creating. As well, India buys the surplus output (after domestic use) with the price based on a cost-plus basis (building in a net return of 15 percent). Other mitigating factors include Bhutan's strong track record of project implementation, rapid growth in energy demand from India, committed donor support, and Bhutan's high level of international reserves.[1] Notwithstanding these mitigating factors, staff cautions against further new non-concessional borrowing as the level of debt has increased substantially compared to previous DSAs. Stress tests to public sector debt dynamics reveal the need for fiscal consolidation and the importance of sustaining rapid economic growth going forward.

[1] Bhutan, with an average Country Policy and Institutional Assessment (CPIA) index of 3.85, is currently classified as a strong performer with regard to its policies and institutions.

BACKGROUND

1. **Bhutan's public and publicly guaranteed (PPG) external debt increased to 85 percent of GDP at end 2012/13 adding 14 percentage points of GDP to the previous year.** The rise in the external public debt was, in large part, driven by hydro sector related external borrowing (with an outstanding stock of 48 percent of GDP). Hydropower projects are primarily financed by India with a mix of loans (70 percent) and grants (30 percent). A large portion of external debt continues to be denominated in Indian rupees (and related to hydropower sector debt), which accounts for 61 percent of total external debt, with convertible currency debt accounting only 29 percent. Domestic debt was entirely due to the loan to purchase aircraft for Druk Air in 2009/10, and the increased domestic debt-to-GDP ratio at about 6 percent of GDP in 2012/13 reflects the fiscal outturns in the following three years.

Bhutan: Structure of Public Sector Debt

	2007/08	2008/09	2009/10	2010/11	2011/12	2012/13
Total Public Sector Debt (percent of GDP)	59	71	57	69	72	90
External	56	69	56	66	71	85
Hydropower sector debt	36	39	32	40	42	48
Non-hydropower sector debt	20	30	24	26	29	37
Domestic	3	2	2	3	1	6

UNDERLYING DEBT SUSTAINABILITY ANALYSIS ASSUMPTIONS

2. **Under the baseline scenario, Bhutan is expected to pursue expansion in its power generation capacity to about 10,000 MW by 2020.** The hydropower development policy of Bhutan consists of eleven new hydropower projects (see text table), which would quintuple Bhutan's power generation capacity. External financing for non-hydropower sector activities continues to remain predominantly from multilateral and bilateral donors at concessional terms. In addition, the government has a stand-by facility with the Government of India to relieve temporary BOP pressures, as well as a swap line with the Reserve Bank of India and overdrafts with Indian commercial banks. These also add to the external debt stock when availed.

Table. Summary of Hydropower Projects in the Pipeline

Project	Capacity (MW)	DPR Schedule	Construction Schedule	Development Model
Punatsangchhu I	1200	Under construction	2008-2017	Intergovernment
Punatsangchhu II	1020	Under construction	2010-2017	Intergovernment
Mangdecchu	720	Under construction	2010-2017	Intergovernment
Sankosh Storage	2650	DPR yet to be cleared	2015-2023	Intergovernment
Kuri Gongri	2640	DPR to be prepared in 11FYP	2017-2026	Intergovernment
Amochhu Storage	540	DPR cleared by CEA	2015-2023	Intergovernment
Wangchhu	570	DPR yet to cleared by CEA, GoI	2015-2023	Joint Venture
Bunakha Storage	180	DPR cleared by CEA	2014-2020	Joint Venture
Kholongchhu	600	DPR approved by two Governments	2014-2021	Joint Venture
Chamkarchhu I	770	DPR yet to be cleared	2015-2024	Joint Venture
Dagachhu	126	Completed	2009-2014	PPP
Total	**11066**			

Source: Annual Report 2012/13, Royal Monetary Authority of Bhutan
Note: CEA: Central Electricity Authority, Government of India.

3. **The hydropower sector developments will govern the rest of the economy as summarized by the following key baseline macroeconomic assumptions**.

- ***Real sector***: Similar to the spike in real GDP when Tala hydropower project was commissioned in 2006/07, Puna I, Puna II and Mangdechu hydropower projects will continue to boost economic growth during the construction phase and as they come on stream in 2017/18. Real growth in the medium-term is projected to remain at around 8½ percent, close to its 10-year historical average of 8 percent. As construction of hydro projects phases out, growth will gradually return to around 6¾ percent in the long-term.
- ***Fiscal sector*:** Upon completion, the hydropower projects will boost the domestic revenue-to-GDP ratio, similar to when Tala hydropower project came on line. External budgetary aid is projected to decline as a share of GDP as Bhutan's per capita income rises. On average, the overall fiscal deficit remains broadly balanced over the long term.
- ***External sector***: The current account deficit is projected to deteriorate over the medium term because of strong growth in import demand associated with the construction phase of the hydropower projects, as well as hydro debt service. Upon the completion of hydropower projects, however, electricity exports are likely to more than triple from current levels, and the current account deficit should decline over the longer term, leading to balance of payment surpluses from 2020 onwards.

Key Macroeconomic Assumptions

	10 year Historical Average	Baseline Average	
		2013–2018	2019–2033
Current transfers, net total (in percent of GDP)	11.3	5.2	0.3
Real GDP growth (percent)	8.0	8.4	6.6
Growth of exports of goods and services (US dollar terms)	21.4	17.6	8.6
Non-interest current account deficit (in percent of GDP)	10.0	15.6	-16.5
Primary deficit (in percent of GDP)	-1.2	-7.1	-11.7

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

A. Baseline

4. **Bhutan's external debt closely traces hydropower cycles**. External debt as a share of GDP is projected to rise to 121 percent of GDP in 2016/17 in line with disbursements for hydropower-sector projects, before declining slightly to 116 percent of GDP in 2017/18 with the completion of the first phase of hydro construction (Table 1). The stock of external public and publicly guaranteed (PPG) debt is expected to gradually decline in the long term to about 29 percent in 2034 once construction of most of the hydropower projects is completed and debt service commences to pay down the debt stock. As a result, the present value (PV) of PPG external debt-to-GDP gradually declines to 25 percent only over the long term. The PV of debt-to-exports ratio falls below the threshold of 200 percent also only in the long run and remains above the threshold for most of the projection period. The PV of PPG external

debt to revenue is also on a downward trend from a peak of 750 percent in 2014/15 and is projected to fall below the threshold of 300 percent only in the long run.[2] However, all the debt ratios remain above the LIC-DSA indicative thresholds for a significant part of the projection period under the baseline. The commissioning of the hydropower projects (in 2017/18), which also marks the start of the debt repayment, puts the debt ratios on a steady downward trajectory.

5. **Debt service-to-export ratio is expected to remain below the indicative thresholds for most of the entire projection period.** However, debt service-to-revenue ratio breaches the indicative threshold as Puna II and Mangdechu's debt service begin in 2018/19; however, it is expected to fall below the threshold over the long run.[3] The protracted breaches of the thresholds over the medium term would indicate a high risk of debt distress, although strong mitigating factors exist for hydropower debt.

6. **Excluding hydropower debt from the total (i.e., treating these loans like FDI), breaches of baseline debt-ratios are temporary and/or marginal**. The assumptions in the scenario are the same as the previous one outlined in paragraph 3. All hydropower loans are excluded to assess debt dynamics of non-hydro debt. The results of this scenario are that non-hydro debt ratios are well below the threshold in case of PV of debt to GDP, exports and revenues. There are temporary breaches in the case of the debt service indicators.

B. Sensitivity Analysis

7. **The PV of debt-to-exports and debt-to-revenue ratios (which are two of the external debt sustainability thresholds) are breached under stress tests** (exports shock, exchange rate depreciation, growth shock and unfavorable financing terms). The standard sensitivity analysis points to a moderate risk of debt distress (Figure 1 and Table 2), as all debt burden indicators (PV of debt-to-exports, debt-to-revenue, and debt-to-GDP ratios) breach their respective thresholds after the shocks in 2014 and only come down in the long run (except under the historical scenario). Based on the bound tests, the most extreme stress for the PV of debt-to-exports or revenue ratios is a one-standard deviation shock to the exchange rate in 2013–14. However, this shock overestimates Bhutan's debt vulnerability since a large share of Bhutan's external income is in Indian rupees, which act as a natural hedge to the rupee-denominated debt.

8. **Debt service ratios also breach their respective thresholds for parts of the projection period.** The debt service-to-export ratio breaches the threshold under a growth shock and the debt service-to-revenue ratio breaches the indicative threshold continuously under the shock scenarios.

[2] The PV indicators exceed ratios in nominal terms due to non-concessional nature of outstanding debt (hydropower-project loans, Dungsam-cement-company borrowing, stand-by-credit-facility with government of India, SWAP arrangement with RMA, and borrowing from Indian commercial banks).

[3] Revenues used to pay for amortization are accounted in the budget as repayable in net lending. The DSA template adds these numbers to revenues to better reflect the capacity of the government to service the debt.

PUBLIC DEBT SUSTAINABILITY ANALYSIS

A. Baseline

9. **The baseline public debt dynamics trace closely the path of external debt**. The public debt-to-GDP ratio is expected to rise to 129 percent in 2016/17 and then decline gradually until it reaches 35 percent of GDP by end 2033/34. External financing is projected to shrink as a share of GDP, making room for domestic financial markets to play a larger role in the financing of the development agenda.

B. Sensitivity Analysis

10. **The public debt ratios are projected to be on a declining path over the long term under various stress tests**. A one-time depreciation results in an upward shift in the various public debt indicators without affecting the shape of the debt indicators' path. However, the sensitivity analysis suggests that the debt service ratios are susceptible to extreme shocks, given the bunching of repayments in the outer years. This scenario underscores the importance of containing fiscal deficits to ensure the sustainability of public debt.

STAFF ASSESSMENT

11. **Despite worsening of debt ratios compared to the 2011 DSA, staff maintains the assessment that Bhutan's risk of external debt distress is moderate due to the presence of unique and strong mitigating factors.** These factors are primarily the explicit guarantees from India that cover financial and construction risks for the hydropower projects (Box 1). As well, India buys the surplus output (after domestic use) with the price based on a cost-plus basis (building in a net return of 15 percent). This unique arrangement allows for an exceptional treatment of hydropower loans from India as similar to FDI—i.e., non-debt creating.

- The addition of new hydropower projects leads to a substantial build-up in external debt, relative to the analysis in the 2011 DSA. The policy-related LIC-DSA thresholds are breached for most of the indicators and remain breached for a longer period compared to the previous analysis. As Bhutan's debt carrying capacity only improves in the long run, staff cautions against any additional non-concessional borrowing. Also, Bhutan's (i) concentrated export base, (ii) rupee-reserves mismatch with external debt and the trade structure of the country (and the consequent buildup of debt related to the overdrafts)—see Selected Issues paper on reserve adequacy and reserve management, and (iii) its fiscal stance, leave it vulnerable to export and constant primary-deficit shocks and any shortfalls in aid inflows. These vulnerabilities are confirmed by the stress tests in the DSA which reveals potential vulnerabilities in Bhutan's external debt situation. Additional risks stem from volatile hydro related debt service payments requiring provisioning of rupee reserves. Two episodes of rupee shortage in 2006/2007 and as recently as 2011/12 are partly explained by lumpy debt service payments.

- However, the guarantees from India mitigate risks related hydropower loans, with debt and debt-service ratios significantly lower and breaches of indicative thresholds temporary and/or marginal, once these loans are excluded. These additional projects also confer strong economic dividends in the long-term, boosting real GDP growth and exports.

12. **Furthermore, mitigating factors that were also highlighted in previous assessments remain valid, namely:**

- Bhutan has a strong track record of project implementation.
- Bhutan maintains close economic and political ties with India, which mitigate the commercial risks of these projects. India has been both the main provider of financing for hydropower projects and the main consumer of the projects' electricity output. Going forward, India's sizeable power deficit will continue to support the demand for Bhutan's hydropower.

Box 1. Institutional and Legal Arrangements of Inter-Governmental Hydropower Projects in Bhutan

Intergovernmental hydropower projects in Bhutan are being undertaken under a unique arrangement between India and Bhutan, reflecting close links between the two countries. Under the "10,000 MW by 2020" bilateral cooperation agreement signed on 26th July 2006 and the protocol to the agreement signed on 16th March 2009, the Royal Government of Bhutan (RGoB) and the Government of India (GoI) agreed to develop 5 intergovernmental projects (Chhuka, Kurichhu, Tala which have been commissioned, and Punatsangchhu I, Punatsangchhu II, Mangdechhu which are under construction).

Under the intergovernmental agreement, a detailed project report (DPR) is prepared by a GoI undertaking. The report reviews all technical and financial aspects of the project. After the DPR is final and endorsed by the 2 countries, a Project Agreement between Bhutan (Ministry of Foreign Affairs) and India (Ministry of External Affairs) is signed, which establishes:

- Sole ownership of the project by the RGoB;
- Estimated cost of project, and establishes GoI's sole responsibility for funding (including additional costs);
- Terms of financing (share of grant versus loan financing from India, simple - non capitalized interest rate, amortization period, with the first repayment starting one year from the mean date of commercial operation)
- Setting-up of a project authority for the construction, operation, maintenance and evacuation of the surplus power, with a chairperson appointed by RGoB;
- Handing over of the project by the project authority to the RGoB within 2 years of completion of the project and dissolution of the project authority at the time;
- That the GoI will purchase all the surplus power from the project (over and above domestic use);
- Principle of a "cost plus" approach for the determination of the power rate at the time of commissioning the project. Cost plus includes the costs of the project, financing costs, operation and maintenance charges, (accelerated) depreciation and market conditions and a net return of 15 percent.
- The principle of reviewing the rate every 3 years.

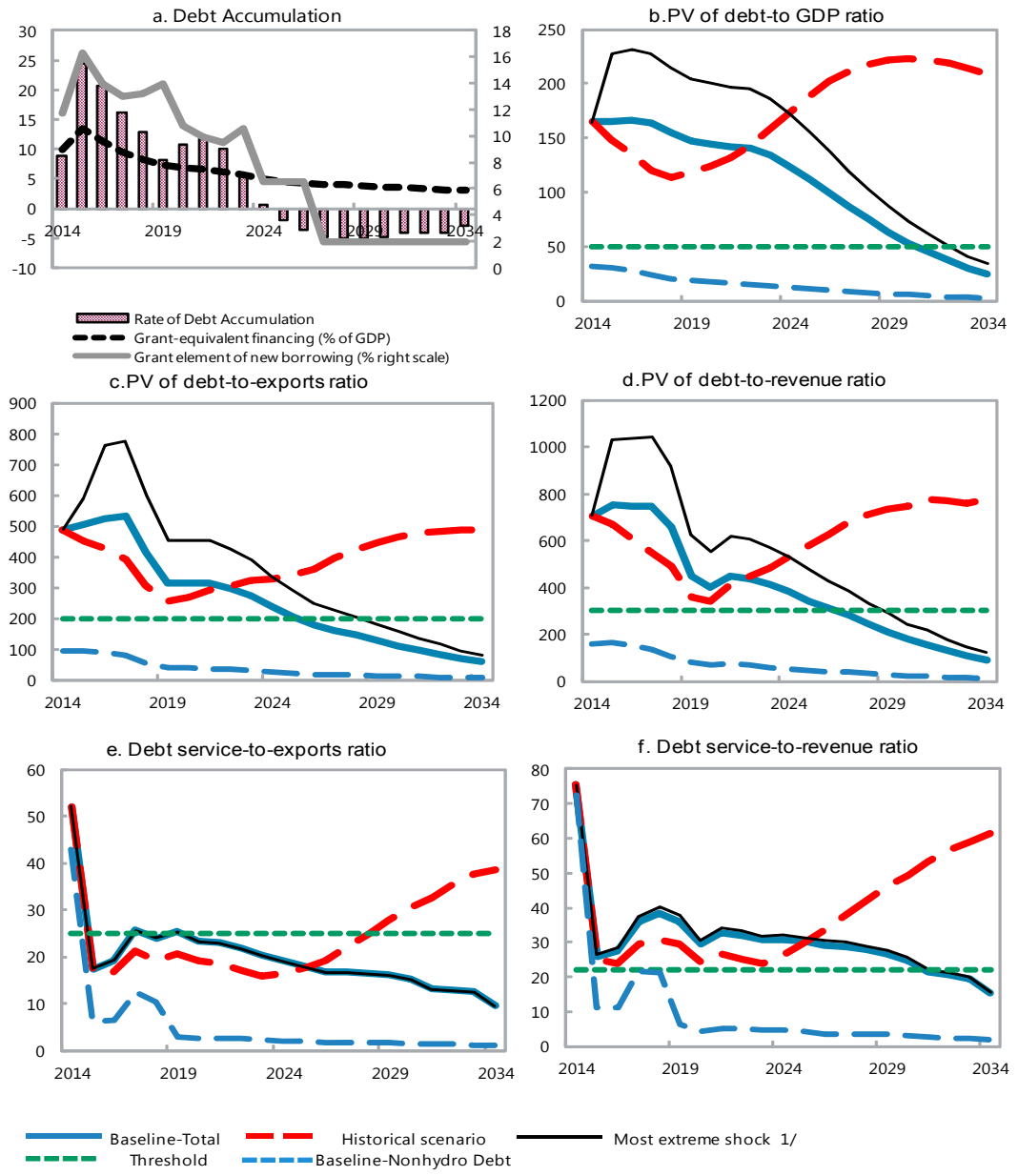
Once the project has finished construction and is commissioned:

- a protocol to the Project Agreement is signed between RGoB (Ministry of Economic Affairs) and GoI (Bhutan Embassy), which establishes the rate at which the surplus power will be sold to GoI, authorizes the designated nodal agencies to enter into a Power Purchasing Agreement (PPA) following the principles established in the Protocol;
- a protocol to the Project Agreement is signed between RGoB (Ministry of Finance) and GoI (Bhutan Embassy), which establishes the principal to be repaid, repayment schedule and interest.
- a subsidiary loan agreement between the Government of Bhutan and Druk Green Power Corporation (DGPC), a Public Holding company part of Druk Holding Investments (DHI), treating the MoF loan as on-lending to DGPC. The agreement follows the same terms as the umbrella agreement, DGPC payments to MoF treasury having to be made a week before the due date to GoI.

<u>Assessment of Debt Sustainability risks of Intergovernmental hydropower projects</u>

Debt sustainability risks are minimal: (i) construction risks are born by GoI, (ii) power rates are determined at the time of the project commissioning, when the actual project cost is known, and are set to allow revenues to service debt and a financial return; (iii) hydropower project are insured (and re-insured) for natural disaster. The only risk is hydrological after commissioning of the project, if electricity, once the project is commissioned, could not be generated.

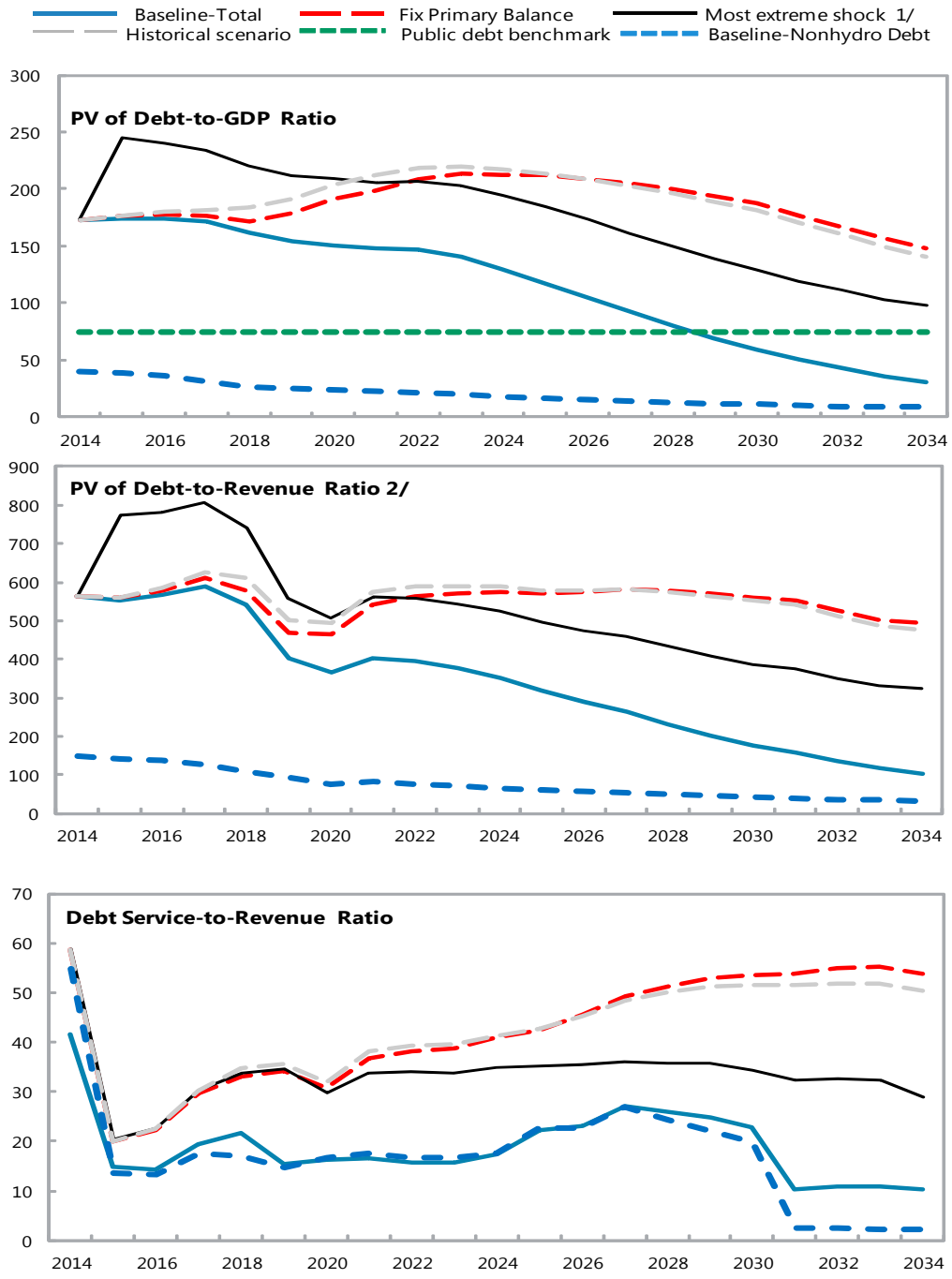
Figure 1. Bhutan: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2014–2034 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Growth shock and in figure f. to a Growth shock

Figure 2. Bhutan: Indicators of Public Debt Under Alternative Scenarios, 2014–2034 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table 1. Bhutan: External Debt Sustainability Framework, Baseline Scenario, 2011–2034 1/

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**65.9**	**71.0**	**84.8**			**105.2**	**114.4**	**120.0**	**120.8**	**115.9**	**116.8**		**117.5**	**29.4**	
of which: public and publicly guaranteed (PPG)	65.9	71.0	84.8			105.2	114.4	120.0	120.8	115.9	116.8		117.5	29.4	
Change in external debt	10.0	5.1	13.8			20.4	9.2	5.6	0.8	-4.9	0.9		-7.5	-5.9	
Identified net debt-creating flows	11.9	15.0	18.4			17.9	17.1	16.5	16.8	1.7	-2.9		-22.3	-13.9	
Non-interest current account deficit	**21.0**	**16.9**	**20.1**	**10.0**	**13.5**	**16.8**	**22.1**	**22.3**	**23.4**	**11.2**	**-2.4**		**-21.7**	**-11.7**	-16.5
Deficit in balance of goods and services	29.7	25.9	23.8			24.8	28.5	29.3	29.2	17.0	7.6		-9.1	-1.9	
Exports	38.1	32.2	31.1			33.8	32.9	31.7	30.6	37.4	46.8		52.4	43.0	
Imports	67.8	58.1	54.9			58.6	61.4	61.0	59.8	54.4	54.4		43.3	41.1	
Net current transfers (negative = inflow)	-12.8	-13.2	-14.1	-11.3	2.9	-4.4	-7.6	-6.7	-5.1	-4.2	-3.0		-1.3	1.8	-0.3
of which: official	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	4.1	4.2	10.4			-3.5	1.1	-0.3	-0.7	-1.6	-7.0		-11.2	-11.6	
Net FDI (negative = inflow)	**-1.5**	**-0.5**	**-1.3**	**-1.5**	**2.1**	**-1.3**	**-1.3**	**-1.3**	**-1.3**	**-1.3**	**-1.3**		**-1.3**	**-1.3**	-1.3
Endogenous debt dynamics 2/	**-7.7**	**-1.4**	**-0.4**			**2.4**	**-3.7**	**-4.5**	**-5.3**	**-8.2**	**0.8**		**0.8**	**-0.9**	
Contribution from nominal interest rate	2.4	2.1	2.9			7.9	3.6	3.9	4.0	4.0	9.2		7.9	1.2	
Contribution from real GDP growth	-4.6	-4.1	-3.4			-5.5	-7.3	-8.4	-9.3	-12.1	-8.4		-7.2	-2.1	
Contribution from price and exchange rate changes	-5.4	0.6	0.0			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-1.8**	**-9.9**	**-4.6**			**2.6**	**-7.9**	**-10.9**	**-16.1**	**-6.7**	**3.8**		**14.8**	**8.0**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	142.5			164.8	165.5	166.1	163.2	154.1	147.0		124.0	24.9	
In percent of exports	...	...	458.3			487.1	503.7	524.6	533.9	411.7	314.1		236.6	57.9	
PV of PPG external debt	**...**	**...**	**142.5**			**164.8**	**165.5**	**166.1**	**163.2**	**154.1**	**147.0**		**124.0**	**24.9**	
In percent of exports	**...**	**...**	**458.3**			**487.1**	**503.7**	**524.6**	**533.9**	**411.7**	**314.1**		**236.6**	**57.9**	
In percent of government revenues	**...**	**...**	**588.2**			**706.9**	**749.4**	**746.4**	**747.5**	**659.0**	**447.9**		**381.7**	**92.5**	
Debt service-to-exports ratio (in percent)	**13.6**	**13.2**	**17.7**			**52.0**	**17.4**	**19.2**	**25.6**	**24.0**	**25.3**		**19.1**	**9.6**	
PPG debt service-to-exports ratio (in percent)	**13.6**	**13.2**	**17.7**			**52.0**	**17.4**	**19.2**	**25.6**	**24.0**	**25.3**		**19.1**	**9.6**	
PPG debt service-to-revenue ratio (in percent)	**23.4**	**18.2**	**22.7**			**75.5**	**25.8**	**27.3**	**35.9**	**38.4**	**36.0**		**30.8**	**15.3**	
Total gross financing need (Billions of U.S. dollars)	0.4	0.4	0.5			0.7	0.5	0.6	0.8	0.6	0.3		-0.6	-1.1	
Non-interest current account deficit that stabilizes debt ratio	11.0	11.8	6.3			-3.6	12.9	16.7	22.6	16.1	-3.3		-14.2	-5.8	
Key macroeconomic assumptions															
Real GDP growth (in percent)	10.1	6.5	5.0	8.0	2.5	6.4	7.6	8.2	8.6	11.5	8.0	8.4	6.2	6.6	6.6
GDP deflator in US dollar terms (change in percent)	10.7	-0.9	0.0	4.8	7.2	-8.6	1.8	2.9	2.8	2.6	2.3	0.6	2.4	2.4	2.4
Effective interest rate (percent) 5/	5.1	3.3	4.3	3.2	1.9	9.1	3.7	3.8	3.7	3.7	8.8	5.5	6.9	3.9	6.8
Growth of exports of G&S (US dollar terms, in percent)	22.2	-10.6	1.2	21.4	30.3	5.7	6.4	7.3	7.8	40.0	38.2	17.6	15.8	6.5	8.6
Growth of imports of G&S (US dollar terms, in percent)	40.5	-9.5	-0.9	20.5	27.5	3.7	14.8	10.6	9.5	4.1	10.5	8.9	3.8	9.1	7.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	11.7	16.3	14.0	13.0	13.2	14.0	13.7	6.5	2.0	5.1
Government revenues (excluding grants, in percent of GDP)	22.1	23.4	24.2			23.3	22.1	22.3	21.8	23.4	32.8		32.5	26.9	30.9
Aid flows (in Billions of US dollars) 7/	0.9	1.4	0.6			0.2	0.3	0.3	0.3	0.3	0.3		0.2	0.4	
of which: Grants	0.2	0.3	0.2			0.1	0.2	0.2	0.2	0.2	0.2		0.2	0.4	
of which: Concessional loans	0.7	1.1	0.4			0.0	0.1	0.1	0.1	0.1	0.1		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			9.9	13.5	11.4	9.4	8.4	7.3		4.9	3.0	4.5
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			33.8	39.1	39.6	38.5	38.9	37.2		52.1	100.0	75.7
Memorandum items:															
Nominal GDP (Billions of US dollars)	1.7	1.8	1.9			1.9	2.1	2.3	2.6	2.9	3.2		5.0	12.0	
Nominal dollar GDP growth	21.9	5.5	4.9			-2.8	9.5	11.3	11.6	14.3	10.5	9.1	8.8	9.2	9.1
PV of PPG external debt (in Billions of US dollars)			2.7			2.9	3.3	3.7	4.1	4.4	4.7		6.1	2.9	
(PVt-PVt-1)/GDPt-1 (in percent)						8.9	24.4	20.6	16.1	12.8	8.2	15.2	0.6	-3.0	-0.2
Gross workers' remittances (Billions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	142.5			164.8	165.5	166.1	163.2	154.1	147.0		124.0	24.9	
PV of PPG external debt (in percent of exports + remittances)	...	...	458.3			487.1	503.7	524.6	533.9	411.7	314.1		236.6	57.9	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	17.7			52.0	17.4	19.2	25.6	24.0	25.3		19.1	9.6	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes. Sizable capital grants are part of residuals.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Bhutan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–2034

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	165	165	166	163	154	147	**124**	25
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	165	148	136	120	114	118	**173**	209
A2. New public sector loans on less favorable terms in 2014-2034 2	165	168	174	175	169	164	**151**	47
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	165	164	170	167	158	151	**127**	25
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	165	165	175	172	162	154	**128**	25
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	165	168	179	176	167	159	**133**	26
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	165	162	164	161	152	145	**122**	24
B5. Combination of B1-B4 using one-half standard deviation shocks	165	162	167	165	156	148	**125**	25
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	165	228	231	227	214	204	**172**	34
PV of debt-to-exports ratio								
Baseline	487	504	525	534	412	314	**237**	58
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	487	451	428	391	304	252	**329**	485
A2. New public sector loans on less favorable terms in 2014-2034 2	487	512	551	573	450	351	**289**	108
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	487	490	514	524	404	308	**231**	55
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	487	588	762	774	596	454	**336**	80
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	487	490	514	524	404	308	**231**	55
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	487	493	518	528	407	310	**232**	55
B5. Combination of B1-B4 using one-half standard deviation shocks	487	487	512	522	403	307	**231**	55
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	487	490	514	524	404	308	**231**	55
PV of debt-to-revenue ratio								
Baseline	707	749	746	748	659	448	**382**	93
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	707	672	609	548	487	360	**531**	775
A2. New public sector loans on less favorable terms in 2014-2034 2	707	762	784	802	721	501	**466**	173
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	707	743	765	767	676	459	**390**	92
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	707	749	788	788	693	471	**393**	93
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	707	760	806	808	712	484	**410**	97
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	707	733	737	739	651	442	**375**	89
B5. Combination of B1-B4 using one-half standard deviation shocks	707	733	753	755	666	452	**385**	91
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	707	1033	1038	1040	917	623	**529**	125

Table 2. Bhutan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–2034 (continued)

Debt service-to-exports ratio								
Baseline	52	17	19	26	24	25	**19**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	52	17	17	21	19	21	**16**	38
A2. New public sector loans on less favorable terms in 2014-2034 2	52	17	17	25	24	26	**23**	16
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	52	17	19	26	24	25	**19**	9
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	52	20	27	38	35	36	**28**	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	52	17	19	26	24	25	**19**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	52	17	19	26	24	25	**19**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	52	17	19	26	24	25	**19**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	52	17	19	26	24	25	**19**	9
Debt service-to-revenue ratio								
Baseline	76	26	27	36	38	36	**31**	15
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	76	25	24	29	31	29	**26**	61
A2. New public sector loans on less favorable terms in 2014-2034 2	76	26	25	35	39	37	**37**	25
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	76	26	29	37	40	38	**32**	16
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	76	26	28	38	40	37	**33**	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	76	27	30	39	42	40	**34**	17
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	76	26	27	36	39	36	**31**	15
B5. Combination of B1-B4 using one-half standard deviation shocks	76	26	28	37	40	37	**31**	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	76	37	39	51	54	51	**43**	21
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	4	4	4	4	4	4	**4**	4

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Bhutan: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–2034

	Actual					Estimate	Projections								
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	68.5	72.3	90.4			113.5	123.3	128.6	128.7	123.1	123.7		123.4	35.3	
of which: foreign-currency denominated	65.9	71.0	84.8			105.2	114.4	120.0	120.8	115.9	116.8		117.5	29.4	
Change in public sector debt	11.1	3.8	18.0			23.1	9.8	5.3	0.2	-5.6	0.6		-7.6	-5.8	
Identified debt-creating flows	-6.6	2.9	-6.6			13.5	-16.4	-14.4	-14.5	-19.8	-20.3		-17.2	-4.7	
Primary deficit	-0.2	-1.2	-2.8	-1.2	2.9	-2.8	-4.5	-4.7	-5.0	-7.7	-18.0	-7.1	-14.6	-2.6	-11.7
Revenue and grants	35.3	37.9	33.5			30.7	31.5	30.8	29.0	29.8	38.1		37.1	30.0	
of which: grants	13.1	14.5	9.3			7.4	9.4	8.5	7.1	6.4	5.3		4.6	3.0	
Primary (noninterest) expenditure	35.1	36.7	30.7			27.9	27.0	26.1	24.0	22.1	20.2		22.5	27.4	
Automatic debt dynamics	-6.4	4.1	-3.9			16.3	-11.9	-9.7	-9.5	-12.2	-2.3		-2.6	-2.2	
Contribution from interest rate/growth differential	-2.8	-7.6	-5.5			-2.5	-8.7	-9.4	-10.1	-13.1	-3.7		-3.9	-2.5	
of which: contribution from average real interest rate	2.5	-3.4	-2.1			2.9	-0.8	-0.1	0.1	0.2	5.4		3.8	0.0	
of which: contribution from real GDP growth	-5.3	-4.2	-3.4			-5.4	-8.0	-9.3	-10.2	-13.3	-9.1		-7.7	-2.6	
Contribution from real exchange rate depreciation	-3.7	11.7	1.6			18.8	-3.2	-0.2	0.6	0.9	1.4		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	17.7	0.9	24.7			9.6	26.2	19.7	14.7	14.2	20.9		9.6	-1.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	148.1			173.0	174.3	174.6	171.1	161.3	153.9		130.0	30.9	
of which: foreign-currency denominated	...	...	142.5			164.8	165.5	166.1	163.2	154.1	147.0		124.0	24.9	
of which: external	...	...	142.5			164.8	165.5	166.1	163.2	154.1	147.0		124.0	24.9	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	5.0	3.7	3.2			14.9	0.2	-0.3	0.6	-1.2	-12.0		-8.1	0.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	442.3			563.9	553.0	567.1	590.5	541.9	403.6		350.8	103.0	
PV of public sector debt-to-revenue ratio (in percent)	...	...	611.2			742.2	789.5	784.8	783.7	689.9	468.9		400.0	114.7	
of which: external 3/	...	...	588.2			706.9	749.4	746.4	747.5	659.0	447.9		381.7	92.5	
Debt service-to-revenue and grants ratio (in percent) 4/	14.7	13.1	17.9			41.6	15.0	14.4	19.4	21.8	15.6		17.3	10.4	
Debt service-to-revenue ratio (in percent) 4/	23.5	21.2	24.7			54.8	21.4	19.9	25.7	27.7	18.1		19.8	11.6	
Primary deficit that stabilizes the debt-to-GDP ratio	-11.3	-5.0	-20.8			-25.9	-14.3	-10.0	-5.1	-2.0	-18.5		-6.9	3.3	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	10.1	6.5	5.0	8.0	2.5	6.4	7.6	8.2	8.6	11.5	8.0	8.4	6.2	6.6	6.6
Average nominal interest rate on forex debt (in percent)	5.1	3.3	4.3	3.2	1.9	12.6	6.0	5.4	5.0	4.8	9.7	7.3	7.3	4.0	7.1
Average real interest rate on domestic debt (in percent)	1.0	3.5	-6.1	1.5	7.4	-44.1	-34.5	-23.5	-19.6	-17.6	-15.8	-25.9	-9.4	-2.3	-6.8
Real exchange rate depreciation (in percent, + indicates depreciation)	-6.9	20.0	2.5	3.3	14.7	...	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	4.6	1.1	14.5	5.8	7.3	2.3	13.4	7.6	6.3	5.9	5.5	6.8	5.7	5.7	5.7
Growth of real primary spending (deflated by GDP deflator, in percent)	-4.7	11.4	-12.2	-0.5	5.7	-3.4	4.1	4.4	-0.1	2.8	-1.3	1.1	5.9	7.0	8.9
Grant element of new external borrowing (in percent)	...	...	...	...	...	11.7	16.3	14.0	13.0	13.2	14.0	13.7	6.5	2.0	...

Sources: Country authorities; and staff estimates and projections.

1/ Gross government debt including hydro-related liabilities.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Bhutan: Sensitivity Analysis for Key Indicators of Public Debt 2014–2034

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	173	174	175	171	161	154	130	31
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	173	177	181	182	183	192	217	140
A2. Primary balance is unchanged from 2014	173	176	178	177	172	179	213	147
A3. Permanently lower GDP growth 1/	173	175	177	175	167	161	152	76
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	173	178	184	182	173	167	153	65
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	173	180	186	183	172	165	145	51
B3. Combination of B1-B2 using one half standard deviation shocks	173	180	188	184	174	167	150	58
B4. One-time 30 percent real depreciation in 2015	173	244	241	234	221	212	194	98
B5. 10 percent of GDP increase in other debt-creating flows in 2015	173	184	184	180	170	163	143	50
PV of Debt-to-Revenue Ratio 2/								
Baseline	564	553	567	591	542	404	351	103
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	564	561	587	627	611	501	588	476
A2. Primary balance is unchanged from 2014	564	558	579	611	577	469	575	492
A3. Permanently lower GDP growth 1/	564	555	573	602	557	421	407	251
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	564	562	591	620	575	434	411	216
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	564	571	605	630	579	432	390	170
B3. Combination of B1-B2 using one half standard deviation shocks	564	570	606	633	583	438	404	194
B4. One-time 30 percent real depreciation in 2015	564	776	781	808	742	556	525	326
B5. 10 percent of GDP increase in other debt-creating flows in 2015	564	583	597	622	571	426	386	166
Debt Service-to-Revenue Ratio 2/								
Baseline	42	15	14	19	22	16	17	10
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	59	20	22	30	35	36	41	50
A2. Primary balance is unchanged from 2014	59	20	22	30	33	34	41	54
A3. Permanently lower GDP growth 1/	59	20	22	30	33	34	35	31
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	59	20	23	30	34	35	35	29
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	59	20	23	31	34	34	34	26
B3. Combination of B1-B2 using one half standard deviation shocks	59	20	23	31	34	35	35	28
B4. One-time 30 percent real depreciation in 2015	59	26	34	45	50	50	51	45
B5. 10 percent of GDP increase in other debt-creating flows in 2015	59	20	23	31	34	34	34	25

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.